

1 Overview

I plan to split this guide into multiple sections: a section for the Geometric Series, Convergent and Divergent Series, Testing for Convergence, Useful Math Tricks, and Solutions to Problems.

This guide is meant to help walk you through each step for a few examples from each section. Within that, there will be detailed solutions to the problems upon request, i.e., email me or review during sessions. I suggest you start without looking at the answers and try your best before looking it up. I'd rather spend hours working on the questions with no references than just googling the answer. This is because it allows you to highlight the gaps in your knowledge. If you need help finding the answer, email me. Do not just look up the answer!

Disclaimer:

This is not an official study guide that Prof. Huff or the Mathematics Department endorsed. It is a compilation of my own previous notes crafted from my personal experience in studying mathematics.

How to Approach Studying Maths:

Breaking the material into manageable sections has been instrumental in my study habits.

If you're in a similar situation as I once was and think that this is impossible and does not make sense, I recommend reaching out to me. We can set up one-on-one sessions if you would like! Do not be afraid to reach out!

Remember, proficiency in maths comes with practice and repetition. However, this doesn't mean you should watch every single YouTube video you see. Instead, focus on your weak points.

How do you identify which questions to prepare for? One effective method is comparing the homework sections with Professor Huff's notes. We all have different approaches to studying, so take control of your learning and find what works best for you.

2 Series Guide

2.1 Geometric Series

An example of a lot of the ideas involved in series is what is known as a geometric series. Within a geometric progression, one multiplies each term by some fixed term so you can get the next term. For example, take the following sequences, which are all geometric progressions.

$$\begin{aligned} 2, 4, 8, 16, 32, \dots, \\ a, ar, ar^2, ar^3, \dots, \end{aligned}$$

Note that the three dots mean that the terms continue as they started, and there is **never** a last term. Thus, the sequence goes on to infinity. If you are asked to find the sum of infinite series, you must realize that not all infinite series have sums. However, even if an infinite series has a finite sum, we cannot always determine it by adding the terms because no matter how many terms we add, there will always be more; like the concept of infinity, how $\infty + 1 = \infty$. A way around this is by defining what it means to be the sum of a series.

Example 1. Derive the formula for the sum S_n of the geometric progression in

$$S_n = a + ar + ar^2 + \dots + ar^{n-1}$$

Step 1: In order to find this general formula for the sum of a general geometric progression, we need to manipulate the expression. We first can multiply r on both sides. After multiplying by S_n by r the expression will become,

$$\begin{aligned} S_n &= a + ar + ar^2 + \dots + ar^{n-1} \\ rS_n &= r(a + ar + ar^2 + \dots + ar^{n-1}) \\ &= ar + ar^2 + ar^3 + \dots + ar^n \end{aligned}$$

Step 2: You then can subtract S_n by rS_n ,

$$\begin{aligned} S_n - rS_n &= (a + ar + ar^2 + \dots + ar^{n-1}) - (ar + ar^2 + ar^3 + \dots + ar^n) \\ &= a - ar^n \end{aligned}$$

Step 3: Factor out like terms,

$$\begin{aligned} S_n - rS_n &= a - ar^n \\ S_n(1 - r) &= a(1 - r^n) \end{aligned}$$

Step 4: Isolate S_n to get the formula for the sum of a finite geometric series,

$$S_n = \frac{a(1 - r^n)}{(1 - r)} \quad (1)$$

Often times the formula above is referred to as the formula for the partial sum of the first n terms in a geometric series.

Definition: The sum of an infinite geometric series if it exists is,

$$S = \lim_{n \rightarrow \infty} S_n \quad (2)$$

S_n is the sum of a finite series, where the series is summed to n terms of the series; often times it is easier to think of S_n as the n th-term in the series. The sum of an infinite geometric series can be found from the previous example by taking the limit as $n \rightarrow \infty$,

$$\begin{aligned} S &= \lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} \frac{a(1 - r^n)}{(1 - r)} \\ &= \lim_{n \rightarrow \infty} \left(\frac{a}{1 - r} - \frac{ar^n}{1 - r} \right) \\ &= \lim_{n \rightarrow \infty} \frac{a}{1 - r} - \lim_{n \rightarrow \infty} \frac{ar^n}{1 - r} \end{aligned}$$

Note: As $n \rightarrow \infty$ then $r^n \rightarrow 0$. Only when $|r| < 1$, but if $|r| > 1$ the result will be ∞ which is why testing for convergence is important! Such that we can conclude that an infinite geometric series has a sum if and only if $|r| < 1$:

$$S = \frac{a}{1-r} \begin{cases} a = \text{first term of the series} \\ r = \text{common ratio} \end{cases} \quad (3)$$

Example 2. Write $0.444444\dots$ as a fraction.

Step 1: Find a and r , and plug them into equation (3). We know to use equation (3) instead of (2) because the three dots imply there is never a last term.

$$0.444444 = \frac{4}{10} + \frac{4}{100} + \frac{4}{1000} + \dots = \frac{4/10}{1 - \frac{1}{10}} = \frac{4}{9}$$

Problem 1. Use equation (3) to find the fractions that are equivalent to the following decimals:

(a) $0.555555\dots$

(b) $0.61111\dots$

Example 3. For further practice with evaluating limits,

$$\lim_{n \rightarrow \infty} \frac{n^2 + 5n^3}{2n^3 + 3\sqrt{4 + n^6}}$$

Step 1: Simplify the expression,

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{n^2 + 5n^3}{2n^3 + 3\sqrt{4 + n^6}} \\ \lim_{n \rightarrow \infty} \frac{n^{-1} + 5}{2 + \frac{3\sqrt{4+n^6}}{n^3}} \end{aligned}$$

Step 2: To make life easier take the limit of the numerator and denominator separately.

$$\begin{aligned} \lim_{n \rightarrow \infty} n^{-1} + 5 &= 5 \\ \lim_{n \rightarrow \infty} 2 + \frac{3\sqrt{4 + n^6}}{n^3} &= 2 + 3 = 5 \\ \lim_{n \rightarrow \infty} \frac{n^2 + 5n^3}{2n^3 + 3\sqrt{4 + n^6}} &= \frac{5}{5} = 1 \end{aligned}$$

2.2 Convergent and Divergent Series

Previously, we were going over series with a finite sum. However, this is only sometimes the case. A series with a finite sum is **CONVERGENT**. If the series does not have a finite sum, it is **DIVERGENT**.

In general, infinite series follow this form $S_n = a_1 + a_2 + a_3 + \dots + a_n + \dots$ each a_n has a unique positive integer n , which consists of numbers or functions given by some formula or rule. (If you are curious how you say this formally $a_n : n \in \mathbb{Z}^+$. If you do not know $\mathbb{Z}^+$ = math symbol for positive integers = $\{1, 2, 3, 4, 5, \dots\}$)

$$\begin{aligned} S_1 &= a_1 \\ S_2 &= a_1 + a_2 \\ S_3 &= a_1 + a_2 + a_3 \\ &\dots \\ S_n &= a_1 + a_2 + a_3 + \dots + a_n \end{aligned}$$

Each S_n is a "partial sum", it is the sum of the first n terms in the series. For each n , S_n stops at what is called the n th term. Recall equation (2),

$$S = \lim_{n \rightarrow \infty} S_n$$

- (a) If the partial sums S_n of an infinite series tend to a limit S , then the series is convergent. Else, it is divergent.
- (b) The limiting value S is called the sum of the series.
- (c) The difference $R_n = S - S_n$ is called the remainder after the n terms. Which implies,

$$\lim_{n \rightarrow \infty} R_n = \lim_{n \rightarrow \infty} (S - S_n) = S - S = 0$$

3 Testing for Convergence

3.1 Preliminary test

Definition: If the terms of an infinite series do not tend to zero such that if $\lim_{n \rightarrow \infty} a_n \neq 0$, the series diverges. If $\lim_{n \rightarrow \infty} a_n = 0$, we must test further.

This is **NOT** a test for convergence but it does weed out hard divergent series.

Example 4. *Given the following series decide whether the series is divergent or requires further testing*

$$\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \dots$$

Step 1: To start recognize that $a_1 = \frac{1}{2}$, such that you then need to find the n th-term of a_n .

$$\frac{1}{2} + \frac{2}{3} + \frac{3}{4} + \frac{4}{5} + \dots + \frac{n}{n+1}$$

Step 2: Take the limit as $n \rightarrow \infty$ for the n th-term,

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = 1$$

Step 3: Conclusion, as the terms tend to 1, by the preliminary test, the series diverges and no further testing is needed.

Problem 2. *Given the following series decide whether the series is divergent or requires further testing. Remember the preliminary test can **NOT** determine whether the series converges!*

- (a) $\sum_{n=1}^{\infty} \frac{1}{n}$
- (b) $\sum_{n=1}^{\infty} (1 - \frac{1}{n^2})$
- (c) $\sum_{n=1}^{\infty} \frac{1}{n!}$
- (d) $\sum_{n=1}^{\infty} \ln(n)$

3.2 Comparison Test

This test has two components, which we can call (a) and (b) respectively.

- (a) Given a series you are testing, $a_1 + a_2 + a_3 + \dots$ we can then let $m_1 + m_2 + m_3 + \dots$ be a series of positive terms which you know converges. Thus it can be concluded that the series $a_1 + a_2 + a_3 + \dots$ is absolutely convergent if $|a_n| \leq m_n$.
- (b) Let $d_1 + d_2 + d_3 + \dots$ be a series of terms which you know diverges. Then the series $|a_1| + |a_2| + |a_3| + \dots$ diverges if $|a_n| \geq d_n$

Example 5. Given the series, test for convergence using the comparison test.

$$\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

Step 1: Write the sum in terms of a_n . Such that $a_n = \frac{1}{\sqrt{n}}$

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$$

Step 2: We then want to find a series that is similar to our a_n . This way we can use the comparison test. A hint here is thinking about what is larger n or $\sqrt{n}$? From this we can compare a_n to $\frac{1}{n}$.

$$a_n > \frac{1}{n}$$

Step 3: Interpret results, as we know $\frac{1}{\sqrt{n}} > \frac{1}{n}$ and if you did problem 2(a) you would know the series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

Step 4: Conclusion, as $|a_n| \geq \frac{1}{n}$ and we know that $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent. Therefore, by the comparison test, $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ also diverges.

Problem 3. Test the following series for convergence using the comparison test.

(a) $\sum_{n=1}^{\infty} \frac{1}{n!}$

(b) $\sum_{n=1}^{\infty} \frac{1}{2^n + 3^n}$

3.3 The Ratio Test

For most cases, the ratio test is another test that for the majority of the time, you will get a result. Recall the geometric series; we were able to determine each term by multiplying the term before it by some ratio r , explicitly saying that $a_n = ra_{n-1}$ or $a_{n+1} = ra_n$, etc. We can find the ratio r from that logic by simply dividing the terms. ie. $r = \frac{a_{n+1}}{a_n}$ etc.

For some cases, the ratio is not constant but will depend on n . This is why we take the absolute value of this ratio. We also find the series limit as $n \rightarrow \infty$; often, this limit is called ρ .

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad (4)$$

For me, when I look at this general formula (4), I find it to be too chaotic, which is why I tend to let $\rho_n = \left| \frac{a_{n+1}}{a_n} \right|$. Such that the equation becomes,

$$\rho_n = \left| \frac{a_{n+1}}{a_n} \right| \quad (5)$$

$$\rho = \lim_{n \rightarrow \infty} \rho_n \quad (6)$$

Recall from example one where we found that a geometric series converges if $|r| < 1$, such that one may think, well, if that is the case, then that implies if $\rho < 1$, then ρ also converges; this is true. Similarly, if $\rho > 1$, it also diverges. If $\rho = 1$, the ratio tells us nothing, and we need to test further.

$$\text{If } \begin{cases} \rho < 1, & \text{the series converges} \\ \rho > 1, & \text{the series diverges} \\ \rho = 1, & \text{needs further testing} \end{cases} \quad (7)$$

Example 6. Use the ratio test to find whether the following series converge or diverge.

$$\sum_{n=1}^{\infty} \frac{2^n}{n^2}$$

Step 1: Realize that $a_n = \frac{2^n}{n^2}$ such that we then need to find our a_{n+1} . In order to find a_{n+1} simply put $(n+1)$ where we see n .

$$\begin{aligned} \text{if } a_n &= \frac{2^n}{n^2} \\ \implies a_{n+1} &= \frac{2^{(n+1)}}{(n+1)^2} \end{aligned}$$

Step 2: We then want to find that "ratio" I had mentioned earlier. The way to do this is by dividing a_{n+1} by a_n . Ensure you then take the absolute value (This is important when dealing with alternating series). Recall that this ratio is what I call ρ_n .

$$\begin{aligned} \rho_n &= \left| \frac{a_{n+1}}{a_n} \right| \\ &= \left| \frac{\frac{2^{(n+1)}}{(n+1)^2}}{\frac{2^n}{n^2}} \right| = \left| \frac{2^{(n+1)}}{(n+1)^2} \cdot \frac{n^2}{2^n} \right| = \left| \frac{2^n \cdot 2}{n^2(1 + \frac{1}{n^2})} \cdot \frac{n^2}{2^n} \right| = \left| \frac{2}{(1 + \frac{1}{n^2})} \right| \\ \implies \rho_n &= \left| \frac{2}{(1 + \frac{1}{n^2})} \right| \end{aligned}$$

Step 3: Now that we have are ratio ρ_n , we then can take the limit of ρ_n as $n \rightarrow \infty$.

$$\begin{aligned} \rho &= \lim_{n \rightarrow \infty} \rho_n \\ &= \lim_{n \rightarrow \infty} \left| \frac{2}{(1 + \frac{1}{n^2})} \right| = \left| \frac{2}{(1 + \frac{1}{\infty})} \right| \\ &= \frac{2}{1 + 0} = \frac{2}{1} = 2 \\ \implies \rho &= 2 \end{aligned}$$

Step 4: In conclusion, if we look at equation (7) and see that as $\rho = 2$, we can then say that as $\rho > 1$, the series diverges.

Problem 4. Use the ratio test to find whether the following series converge or diverge.

(a) $1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \dots + \frac{1}{n!} + \dots$

(b) $\sum_{n=1}^{\infty} \frac{1}{n}$

(c) $\sum_{n=1}^{\infty} \frac{5^n (n!)^2}{(2n)!}$

(d) $\sum_{n=1}^{\infty} \frac{(2n!)}{5n+1}$

3.4 Helpful Math Tricks

Factorial

$$n! = n \cdot (n-1) \cdot (n-2) \cdot (n-3) \cdot \dots \cdot 3 \cdot 2 \cdot 1$$

Example 7.

$$\begin{aligned} 5! &= 5 \cdot (5-1) \cdot (5-2) \cdot (5-3) \cdot (5-4) \\ &= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\ &= 120 \end{aligned}$$

Example 8. Often times you might be asked how to simplify an expression involving factorials.

$$\frac{(n+2)!}{n!}$$

Step 1: Expand $(n+2)!$

$$(n+2)! = (n+2)(n+1)n(n-1)(n-2) \dots 1$$

Step 2: Recognize that $n(n-1)(n-2) \dots 1$ is the same as $n!$ so we can simplify the expression,

$$\frac{(n+2)!}{n!} = \frac{(n+2)(n+1)n(n-1)(n-2) \dots 1}{n!} = \frac{(n+2)(n+1)n!}{n!} = (n+2)(n+1)$$

3.5 Solutions

Solution 1:

- (a) $\frac{5}{9}$
- (b) $\frac{11}{18}$

Solution 2:

- (a) Requires further testing (Only using the preliminary test; we know $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent).
- (b) The series diverges, $\lim_{n \rightarrow \infty} a_n = 1$.
- (c) Requires further testing, $\lim_{n \rightarrow \infty} a_n = 0$.
- (d) The series diverges, $\lim_{n \rightarrow \infty} a_n = \infty$.

Solution 3:

- (a) Compare to this geometric series $b_n = \sum_{n=1}^{\infty} \frac{1}{2^n}$. If we expand this out you will see the terms of $b_n > \frac{1}{n!}$ for all terms $n > 3$. We know $\sum_{n=1}^{\infty} \frac{1}{2^n}$ converges because it is a geometric series with a ratio of $\frac{1}{2}$. Thus $\sum_{n=1}^{\infty} \frac{1}{n!}$ converges also.
- (b) Compare $\sum_{n=1}^{\infty} \frac{1}{2^n+3^n}$ with $\sum_{n=1}^{\infty} \frac{1}{2^n}$, thus $\frac{1}{2^n} > \frac{1}{2^n+3^n}$; similarly as in problem 3a, since $\sum_{n=1}^{\infty} \frac{1}{2^n}$ and the ratio $r < 1$, the series converges.

Solution 4:

- (a) $\rho = \lim_{n \rightarrow \infty} \rho_n = 0$, since $\rho < 1$, the series converges.
- (b) $\rho = \lim_{n \rightarrow \infty} \rho_n = 1$, since $\rho = 1$, we need to test further.
- (c) $\rho = \lim_{n \rightarrow \infty} \rho_n = \frac{5}{4}$, since $\rho > 1$, the series diverges.
- (d) $\rho = \lim_{n \rightarrow \infty} \rho_n = \infty$, since $\rho > 1$, the series diverges.